Recall — the cube and the cuboid

Faces, edges and vertices; volume as a product of three lengths; the litre.

LESSON 139 | 1 × 40 MIN | MATEMATIKA 6, TAKRORLASH | S7 15 DISTANCE, AREA AND VOLUME

LESSON CLOCK

10' | 12' | 12' | 6'

The two solids	10 min
Volume	12 min
Surface area	12 min
Units and capacity	6 min

LEARNING OBJECTIVES

- Name the faces, edges and vertices of a cuboid.
- Find the volume of a cuboid and of a cube.
- Find the total surface area from the three pairs of faces.
- Convert between cm^3, m^3 and litres.

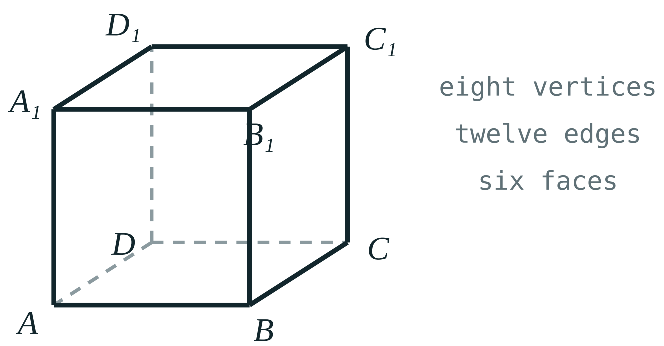
TEXTBOOK

National	pp. 400–403
Cambridge	Stage 7, pp. 156–160
Workbook	Exercise 15.1

01 Explanation

The two solids

A cuboid has six rectangular faces, meeting in twelve edges at eight vertices. A cube is the cuboid whose twelve edges are all the same length.



A cube — six faces, twelve edges, eight vertices

SOLID	FACES	EDGES	VERTICES	FACES ARE
cuboid	*6*	*12*	*8*	rectangles, equal in opposite pairs
cube	*6*	*12*	*8*	six equal squares

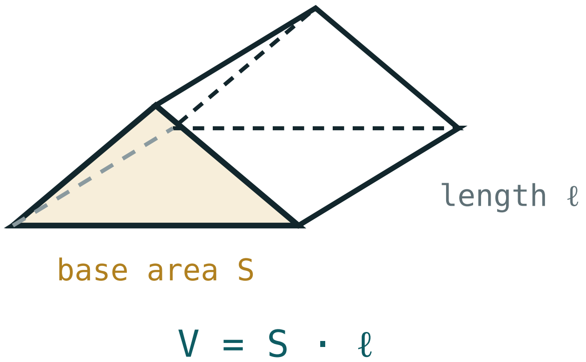
THREE LENGTHS DESCRIBE A CUBOID COMPLETELY

Length, width and height — usually written a, b, c. For a cube all three are the same, so one number is enough.

Volume

> $V = abc$ and for a cube $V = a^3$

The bottom layer holds ab unit cubes, and there are c such layers.



Area of the base multiplied by the height — the rule behind every prism

SOLID	DIMENSIONS	WORKING	VOLUME
cuboid	$5 \times 4 \times 3$ cm	$5 \cdot 4 \cdot 3$	60 cm^3
cuboid	$8 \times 5 \times 2$ cm	$8 \cdot 5 \cdot 2$	80 cm^3
cuboid	$10 \times 6 \times 4$ cm	$10 \cdot 6 \cdot 4$	240 cm^3
cube	4 cm edge	4^3	64 cm^3

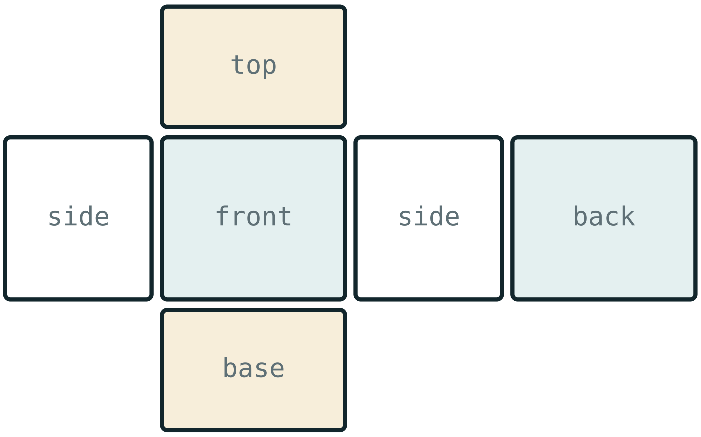
ALL THREE LENGTHS IN THE SAME UNIT

A box 1 m by 50 cm by 40 cm is $100 \times 50 \times 40 = 200\,000$ cm³, not $1 \times 50 \times 40$. Convert first, multiply second.

Surface area

The six faces come in three equal pairs, so the total is twice the sum of three rectangles.

$$S = 2(ab + bc + ac) \text{ and for a cube } S = 6a^2$$



The net — the six faces laid flat, which is what the formula adds up

SOLID	THE THREE RECTANGLES	SUM	SURFACE AREA
$5 \times 4 \times 3$	$20, 12, 15$	47	94 cm²
$8 \times 5 \times 2$	$40, 10, 16$	66	132 cm²
$10 \times 6 \times 4$	$60, 24, 40$	124	248 cm²
cube, edge 4	$16, 16, 16$	48	96 cm²

AREA IS SQUARED, VOLUME IS CUBED

Two lengths multiplied give cm^2; three give cm^3. An answer in the wrong unit is almost always the wrong quantity.

Units and capacity

UNIT	EQUALS	EVERYDAY SIZE
$1\ cm^3$	$1\ ml$	a small sugar cube
$1000\ cm^3$	1 litre	a cube of edge 10 cm
$1\ m^3$	$1\ 000\ 000\ cm^3 = 1000$ litres	a cube of edge 1 m

1 M³ IS A MILLION CM³, NOT A HUNDRED

Each of the three lengths is multiplied by 100, so the volume is multiplied by $100^3 = 1\ 000\ 000$. The same trap turns 1 m² into $10\ 000$ cm², not 100.

02 Worked examples

EXAMPLE 1 **Find the volume and the surface area of a cuboid 5 cm by 4 cm by 3 cm.**

1. $V = 5 \cdot 4 \cdot 3 = 60$ cm³.
2. The three rectangles: $20, 12, 15$.
3. $S = 2 \cdot 47 = 94$ cm².

   Volume in cm^3, area in cm^2 ✓

ANSWER

60 cm³ and 94 cm²

EXAMPLE 2 **A cube has edge *6* cm. Find its volume and surface area.**

1. $V = 6^3 = 216$ cm³.
2. $S = 6 \cdot 6^2 = 6 \cdot 36$.
3. $S = 216$ cm².
   The numbers agree only for edge 6 — a coincidence, not a rule.

ANSWER

216 cm³ and 216 cm²

EXAMPLE 3 **A tank measures *50* cm by *40* cm by *30* cm. How many litres does it hold?**

1. $V = 50 \cdot 40 \cdot 30 = 60\,000$ cm³.
2. 1000 cm³ is 1 litre.
3. $60\,000 \div 1000 = 60$ litres.

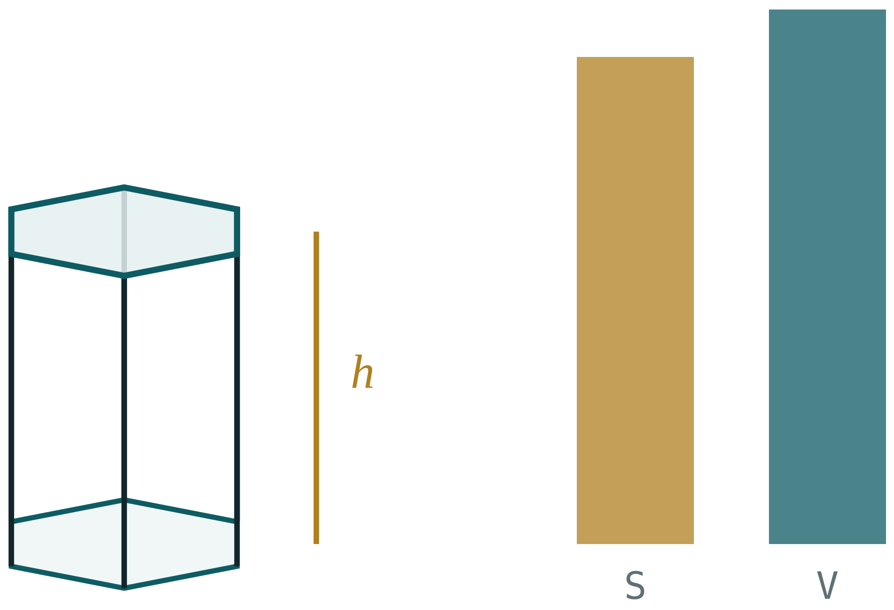
ANSWER

60 litres

03 Interactive model

Hold up a chalk box and ask for the volume and the surface area of the same object; separating “how much fits in” from “how much wraps round” is the whole lesson.

Base area times height



base area B **41.6** | perimeter P **24** | total S **227.1** | volume V **249.4**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Cuboid	Parallelepiped	Прямоугольный параллелепипед
Cube	Kub	Куб
Face	Yoq	Грань
Edge	Qirra	Ребро
Vertex	Uchi	Вершина
Volume	Hajm	Объём
Surface area	Sirt yuzasi	Площадь поверхности
Litre	Litr	Литр

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A cuboid has:

- 4 faces
- 6 faces
- 8 faces
- 12 faces

The volume of a $5 \times 4 \times 3$ cuboid is:

- 12
- 60
- 94
- 47

The surface area of that cuboid is:

- 47
- 60
- 94
- 188

A cube of edge 4 cm has volume:

- 12
- 16
- 64
- 96

1 litre is:

- 100 cm³
- 1000 cm³
- 1 m³
- 10 cm³

1 m³ in cm³ is:

- 100
- $10\,000$
- $1\,000\,000$
- 1000

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. The number of faces of a cuboid

   ANSWER 6

2. The number of edges of a cuboid

   ANSWER 12

3. The number of vertices of a cuboid

   ANSWER 8

4. The volume of a cube of edge 4 cm

   ANSWER 64 cm^3

5. The volume of a $5 \times 4 \times 3$ cm cuboid

   ANSWER 60 cm^3

6. The surface area of a cube of edge 4 cm

   ANSWER 96 cm^2

7. 1 litre in cm^3

   ANSWER 1000 cm^3

Medium · the standard question

7 PROBLEMS

1. The volume of an $8 \times 5 \times 2$ cm cuboid

   ANSWER 80 cm^3

2. The surface area of a $5 \times 4 \times 3$ cm cuboid

   ANSWER 94 cm^2

3. The surface area of an $8 \times 5 \times 2$ cm cuboid

   ANSWER 132 cm^2

4. The volume of a cube of edge 10 cm, in litres

   ANSWER 1 litre

5. The volume of a $10 \times 6 \times 4$ cm cuboid

   ANSWER 240 cm^3

6. The surface area of a cube of edge 5 cm

   ANSWER 150 cm^2

7. 1 m^3 in litres

   ANSWER 1000 litres

Hard · stretch and reason

7 PROBLEMS

1. A cube has volume 216 cm³: its edge

   ANSWER 6 cm

2. …and its surface area

   ANSWER 216 cm²

3. A $12 \times 5 \times h$ cm cuboid has volume 300 cm³: find h

   ANSWER 5 cm

4. 2 m³ in cm³

   ANSWER $2\ 000\ 000$ cm³

5. A tank $50 \times 40 \times 30$ cm, in litres

   ANSWER 60 litres

6. How many cubes of edge 3 cm fill a cube of edge 6 cm?

   ANSWER 8

7. Which cube has surface area equal in number to its volume?

   ANSWER Edge 6, both 216

07 Homework

Homework — 5 tasks

Write the unit on every answer; it is the quickest check that the right quantity was found.

1. Find the volume and the surface area of a cuboid 7 cm by 4 cm by 2 cm.
2. Find the volume and the surface area of a cube of edge 5 cm.
3. A box is 1 m by 60 cm by 50 cm. Find its volume in cm³.
4. How many litres does a tank 80 cm by 50 cm by 25 cm hold?
5. Explain in one sentence why 1 m³ is $1\ 000\ 000$ cm³ and not 100 cm³.